

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
NG_FFCutsFits =
recover_selected_cuts('NG_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_mult
icut_allJ\sel_cuts');

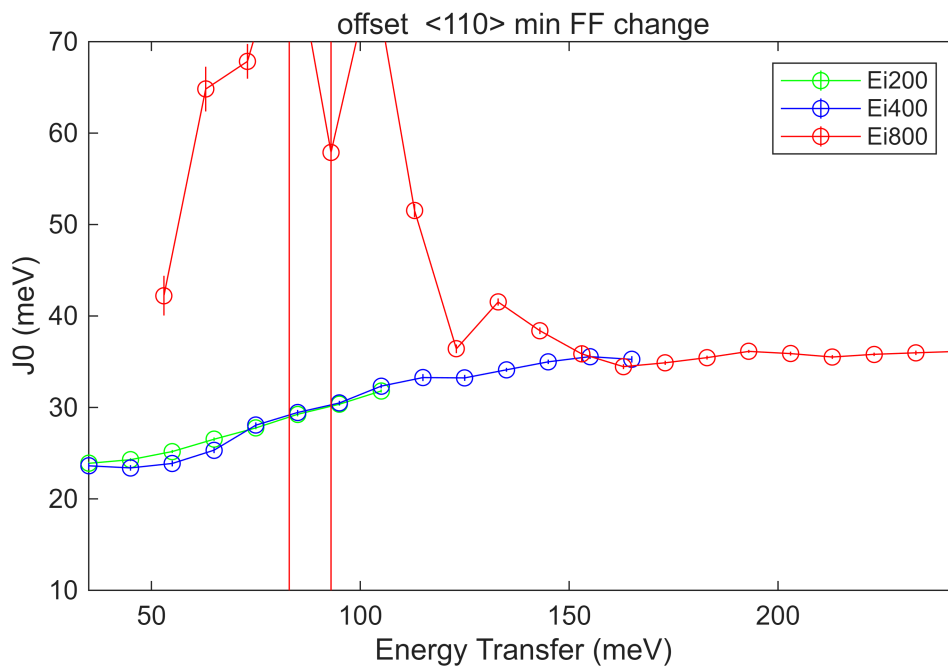
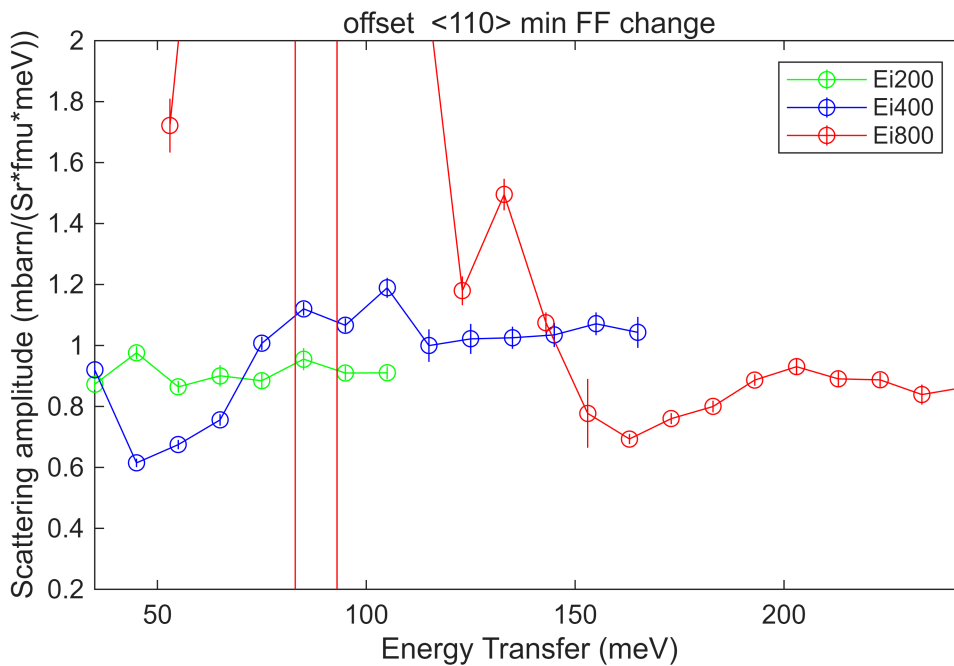
fit_res_field = 'all_fit_par';
separate_fig = false; % if true, detach figures from live script

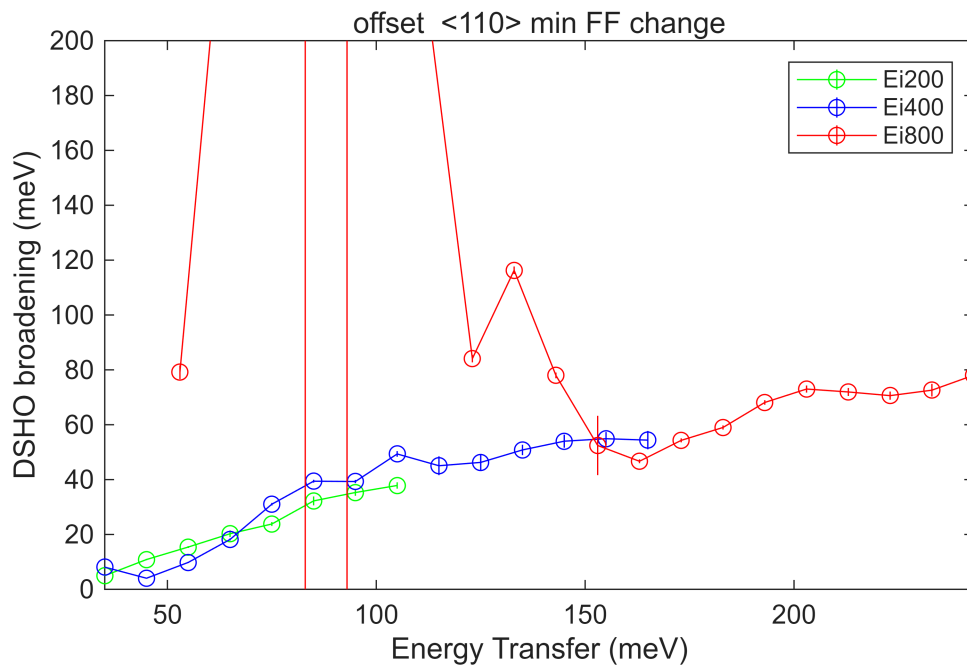
```

```

[S_op110,J_op110,G_op110,Sq_op110,proj110minFF]=
display_EiDep_blocks(NG_FFCutsFits, ...
    'off110NGminFF',' <110> min FF change',fit_res_field,separate_fig);

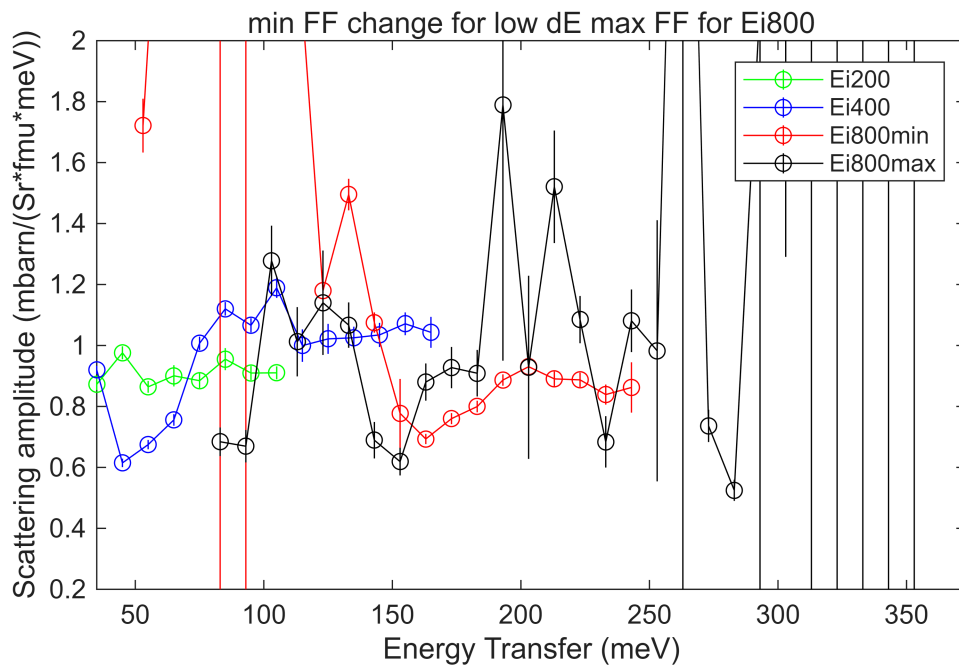
```

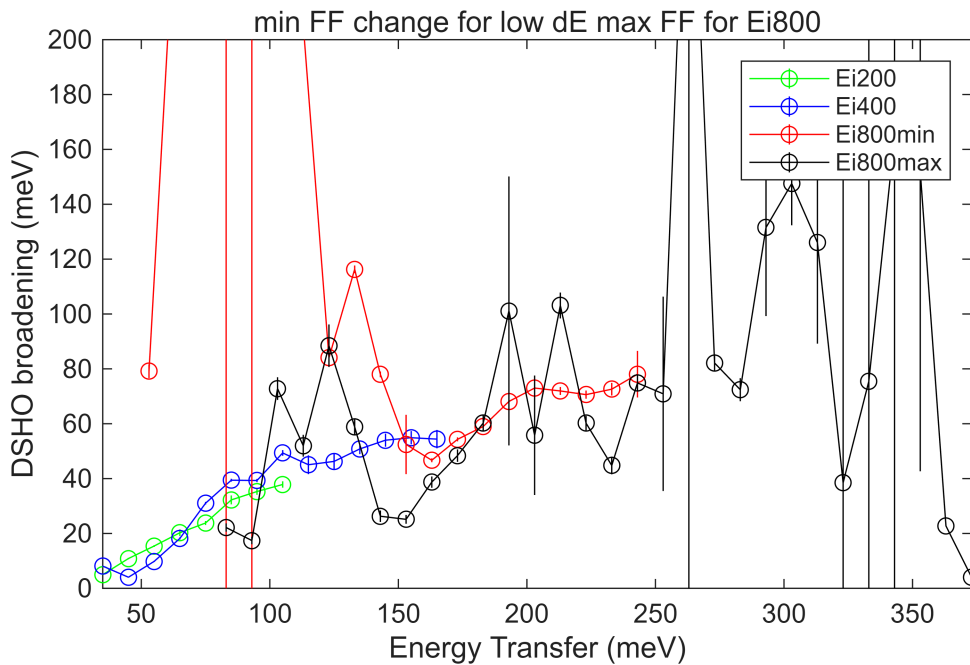
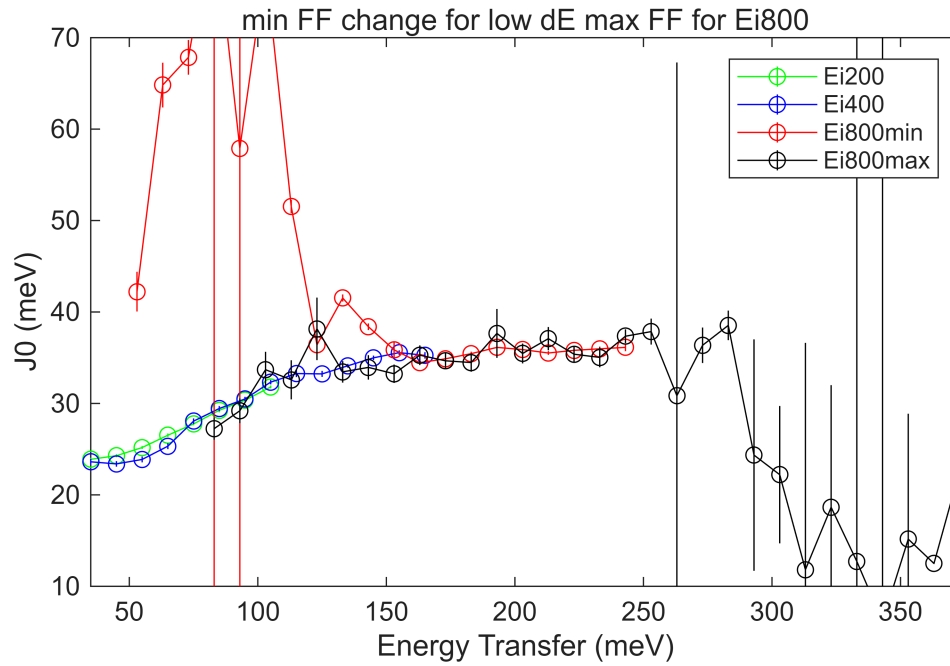




```
[S_maEi800,J_maEi800,Gam_maEi800,S_maEi800q] = extract_fit_par( ...
    NG_FFCutsFits.(['Ei800_off200NGmaxFF_',fit_res_field]));
proj200maxFF = NG_FFCutsFits.Ei200_off200NGmaxFF.data.proj;

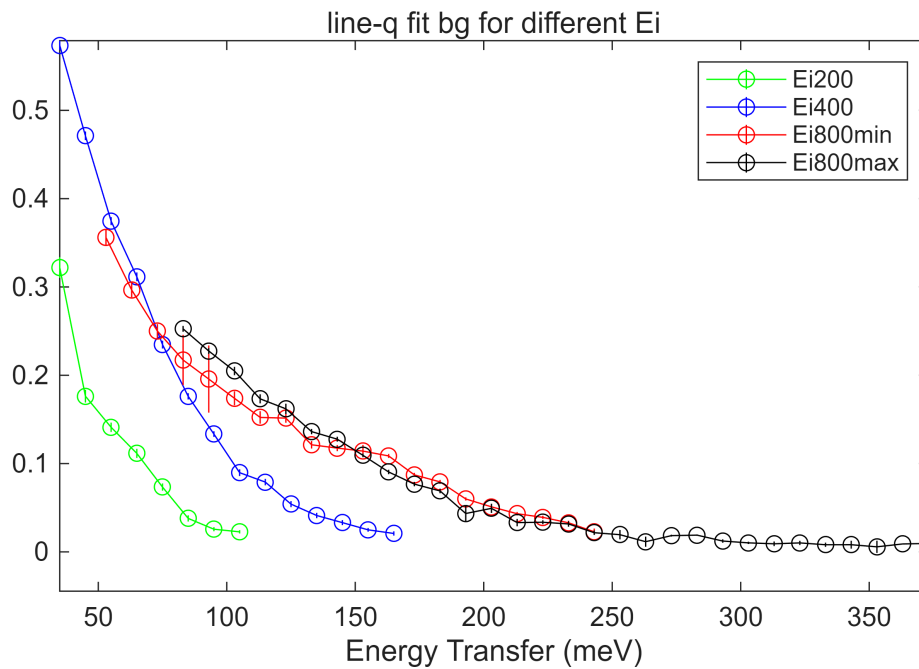
plot_fp(separate_fig,'min FF change for low dE max FF for Ei800', ...
    [S_op110,S_maEi800], ...
    [J_op110,J_maEi800], ...
    [G_op110,Gam_maEi800]);
```





```
bg_dataNGEi200 = extract_bg_par(NG_FFCutsFits.
(['Ei200_off110NGminFF_',fit_res_field]),1);
bg_dataNGEi400 = extract_bg_par(NG_FFCutsFits.
(['Ei400_off110NGminFF_',fit_res_field]),1);
bg_dataNGEi800mi = extract_bg_par(NG_FFCutsFits.
(['Ei800_off110NGminFF_',fit_res_field]),1);
bg_dataNGEi800ma = extract_bg_par(NG_FFCutsFits.
(['Ei800_off200NGmaxFF_',fit_res_field]),1);

plot_fp(separate_fig,'line-q fit bg for different Ei',[], ...
[bg_dataNGEi200,bg_dataNGEi400,bg_dataNGEi800mi,bg_dataNGEi800ma])
```

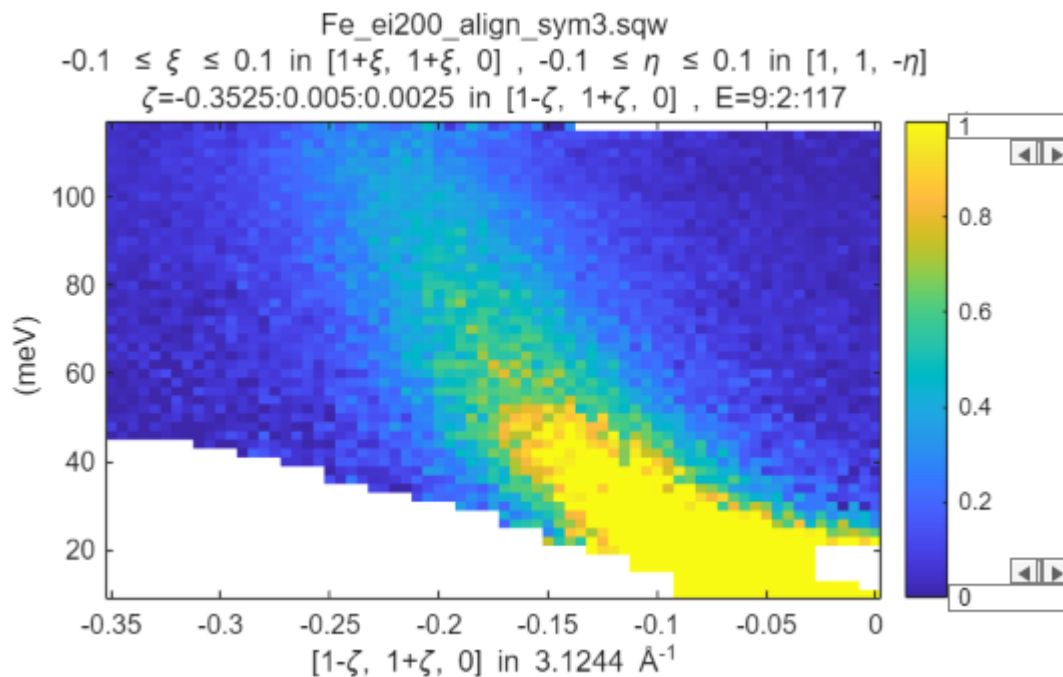


```
spp.Ei200_off110NGminNbg = NG_FFCutsFits.Ei200_off110NGminFF - ...
    sqw_eval(NG_FFCutsFits.Ei200_off110NGminFF,@linterp_bg,bg_dataNGEi200);
spp.Ei200_off110NGminNbg = cut(spp.Ei200_off110NGminNbg, ...
    [-0.35,0.005,0],[],SymopReflection('normvec',[-1,1,0],'offset',[1,1,0]))
```

```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 3913014 pixels -- processing data... -----> retained      28987 pixels
-----> retained 3884027 pixels
-----
Number of points in input object: 3913014
  Fraction of object selected: 100.00% (= 3913014 points) ; Access speed: 1.46GB/sec
  Fraction of object retained: 100.00% (= 3913014 points)

**** Total time in cut(sqw) : 1.3sec Including:
    Data access time fraction: 13.9%; Transf. time frac.: 52.6%; Resulting pix preparation: 11.6%
-----
spp = struct with fields:
  Ei200_off110NGminNbg: [1x1 sqw]
  Ei800_off200NGmaxNbg: [1x1 sqw]
```

```
plot(spp.Ei200_off110NGminNbg); lz 0 1; keep_figure
```



```
spp.Ei400_off110NGminNbg = NG_FFCutsFits.Ei400_off110NGminFF - ...
    sqw_eval(NG_FFCutsFits.Ei400_off110NGminFF,@linterp_bg,bg_dataNGEi400);
spp.Ei400_off110NGminNbg = cut(spp.Ei400_off110NGminNbg,[-0.5,0.005,0],
[,SymopReflection('normvec',[-1,1,0],'offset',[1,1,0]))
```

```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 2187417 pixels -- processing data... -----> retained 11419 pixels
-----> retained 2175998 pixels
```

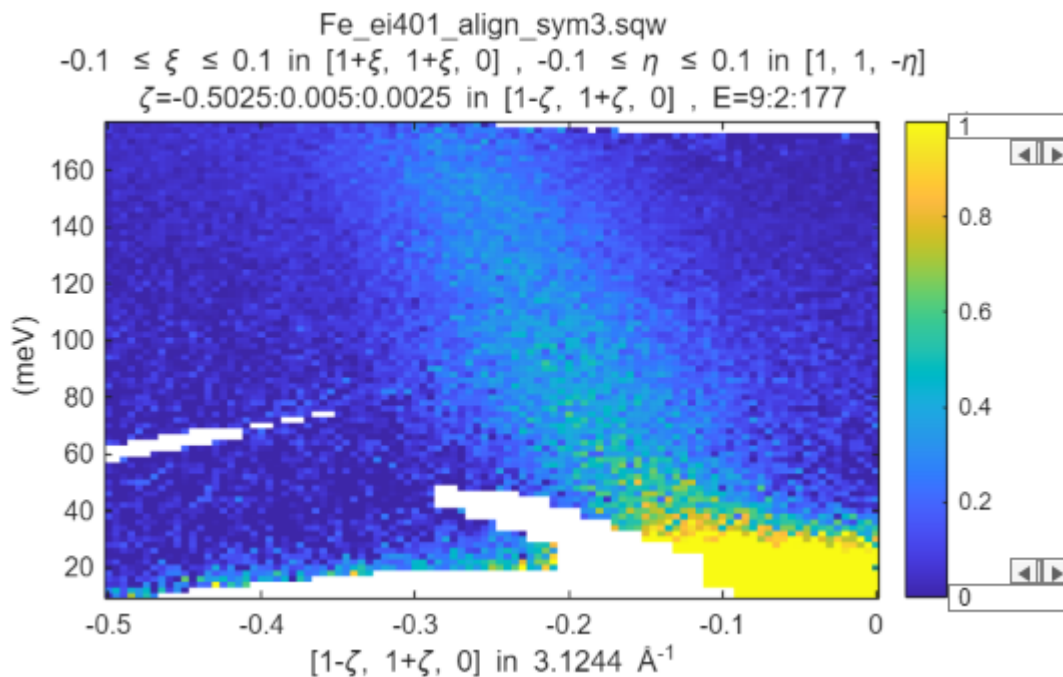
```
-----
Number of points in input object: 2187417
```

```
Fraction of object selected: 100.00% (= 2187417 points) ; Access speed: 1.61GB/sec
Fraction of object retained: 100.00% (= 2187417 points)
```

```
**** Total time in cut(sqw) : 0.7sec Including:
Data access time fraction: 12.2%; Transf. time frac.: 55.6%; Resulting pix preparation: 10.8%
-----
```

```
spp = struct with fields:
Ei200_off110NGminNbg: [1x1 sqw]
Ei800_off200NGmaxNbg: [1x1 sqw]
Ei400_off110NGminNbg: [1x1 sqw]
```

```
plot(spp.Ei400_off110NGminNbg); lz 0 1; keep_figure
```



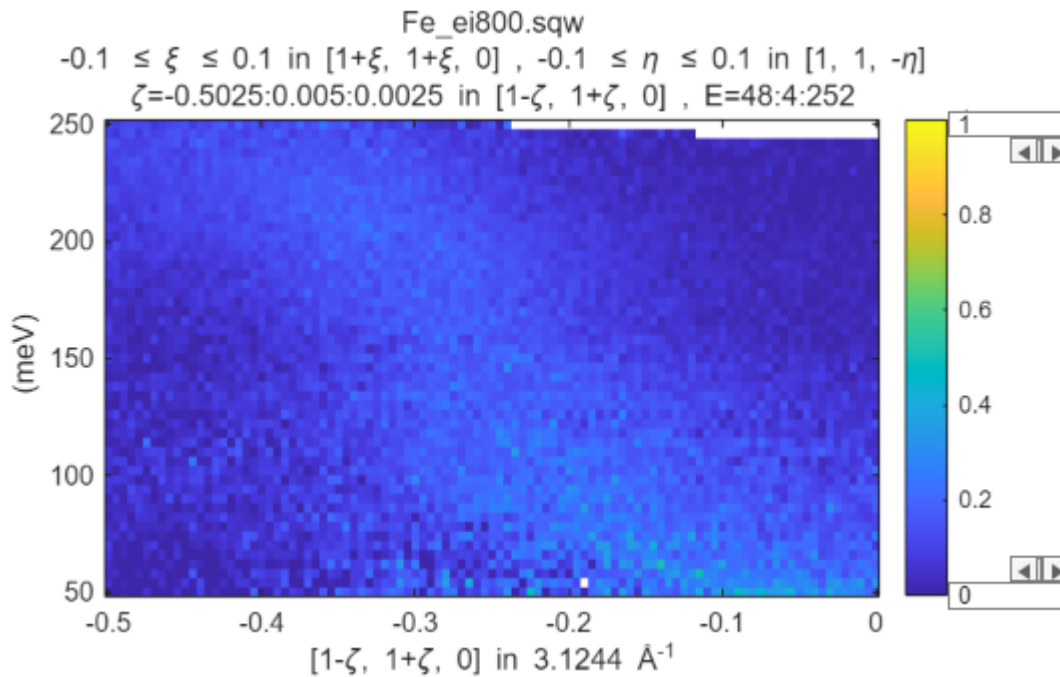
```
spp.Ei800_off110NGminNbg = NG_FFCutsFits.Ei800_off110NGminFF - ...
    sqw_eval(NG_FFCutsFits.Ei800_off110NGminFF,@linterp_bg,bg_dataNGEi800mi);
spp.Ei800_off110NGminNbg = cut(spp.Ei800_off110NGminNbg, ...
    [-0.5,0.005,0],[],SymopReflection('normvec',[-1,1,0],'offset',[1,1,0]));
```

```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 929980 pixels -- processing data... -----> retained 4925 pixels
-----> retained 925055 pixels
```

```
-----
Number of points in input object: 929980
    Fraction of object selected: 100.00% (= 929980 points) ; Access speed: 1.15GB/sec
    Fraction of object retained: 100.00% (= 929980 points)
```

```
**** Total time in cut(sqw) : 0.4sec Including:
    Data access time fraction: 14.6%; Transf. time frac.: 52.6%; Resulting pix preparation: 12.8%
-----
```

```
plot(spp.Ei800_off110NGminNbg); lz 0 1; keep_figure
```



```
spp.Ei800_off200NGmaxNbg = NG_FFCutsFits.Ei800_off200NGmaxFF - ...
    sqw_eval(NG_FFCutsFits.Ei800_off200NGmaxFF,@linterp_bg,bg_dataNGEi800ma);
spp.Ei800_off200NGmaxNbg = cut(spp.Ei800_off200NGmaxNbg, ...
    [-0.5,0.005,0],[,],SymopReflection('normvec',[1,1,0],'offset',[2,0,0]))
```

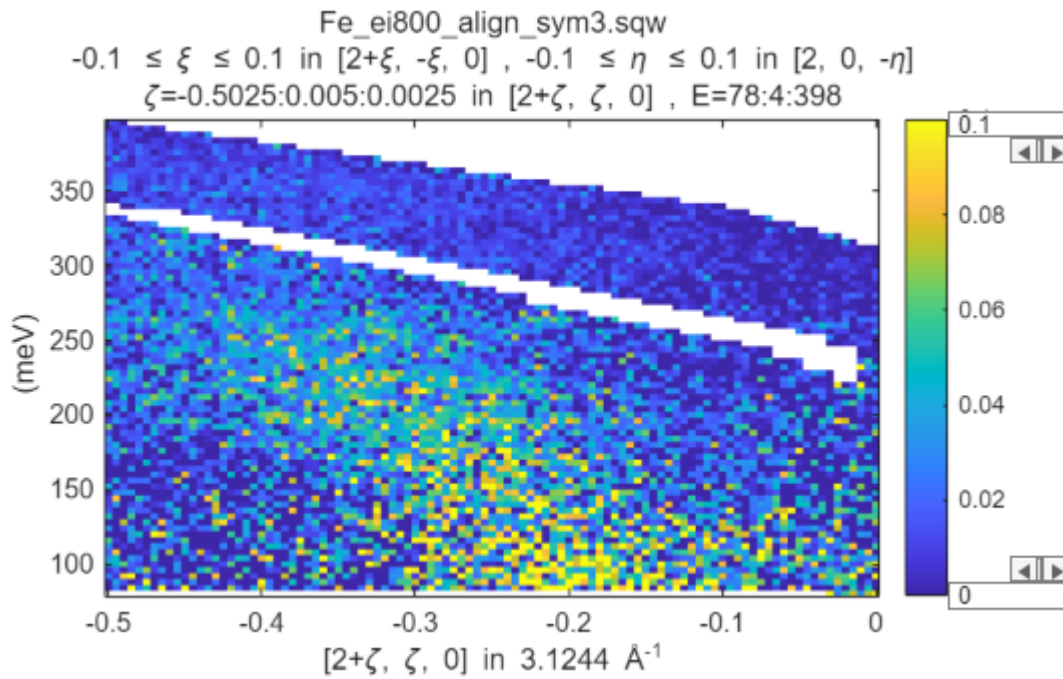
```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 968255 pixels -- processing data... -----> retained 11620 pixels
-----> retained 940479 pixels
```

```
-----
Number of points in input object: 968255
Fraction of object selected: 100.00% (= 968255 points) ; Access speed: 1.48GB/sec
Fraction of object retained: 98.33% (= 952099 points)
```

```
**** Total time in cut(sqw) : 0.4sec Including:
Data access time fraction: 11.2%; Transf. time frac.: 58.8%; Resulting pix preparation: 11.3%
```

```
-----
spp = struct with fields:
Ei200_off110NGminNbg: [1x1 sqw]
Ei800_off200NGmaxNbg: [1x1 sqw]
Ei400_off110NGminNbg: [1x1 sqw]
Ei800_off110NGminNbg: [1x1 sqw]
```

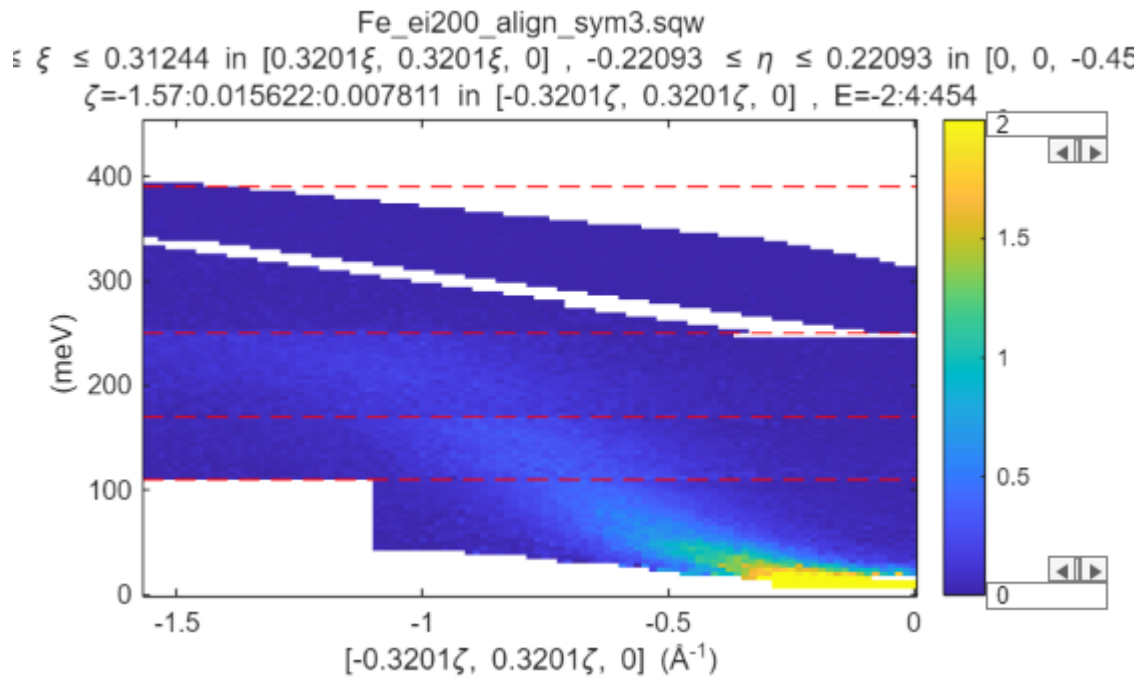
```
plot(spp.Ei800_off200NGmaxNbg); lz 0 0.1; keep_figure
```



```
cuts_rangesNG = containers.Map();
cuts_rangesNG('Ei200_off110NGminNbg') = [0;110];
cuts_rangesNG('Ei400_off110NGminNbg') = [110;170];
cuts_rangesNG('Ei800_off110NGminNbg') = [170;250];
cuts_rangesNG('Ei800_off200NGmaxNbg') = [250;390];

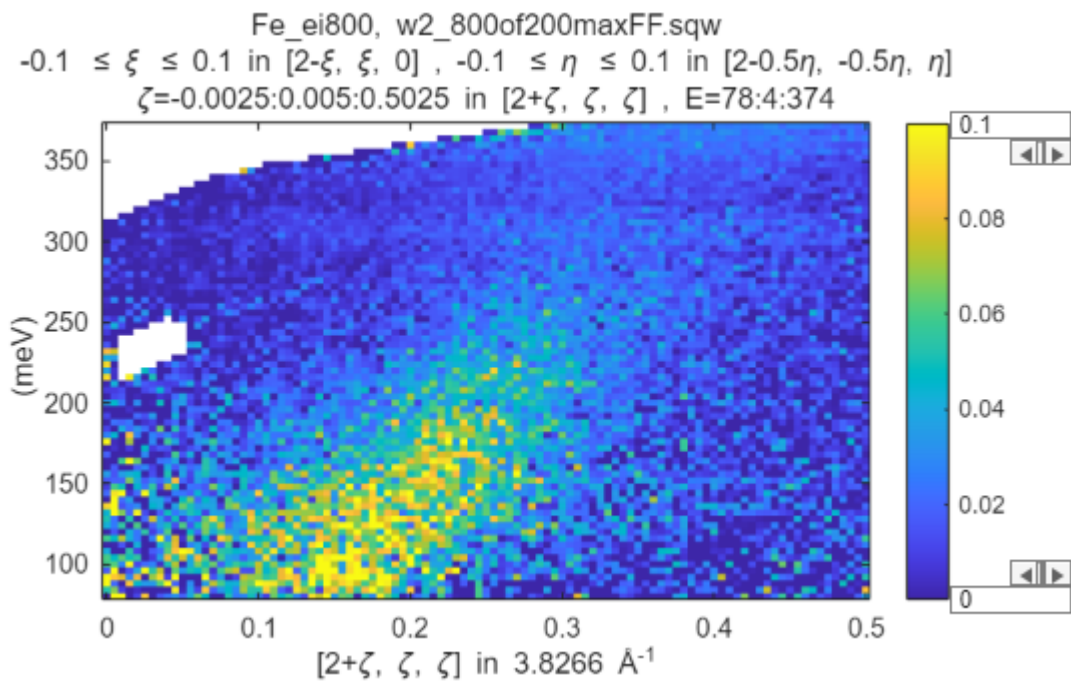
[w2,fh] = build_disp_in_dir_with_ds(spp,cuts_rangesNG,[-0.5,0.005,0],false);
```

```
*** Finished 1of#4 chunks in 1.0323 sec performing sqw_op_bin_pixels
...
*** Completed sqw_op_bin_pixels using 4 pages in 2.1768 sec.
*** Resulting object is backed by file:
```

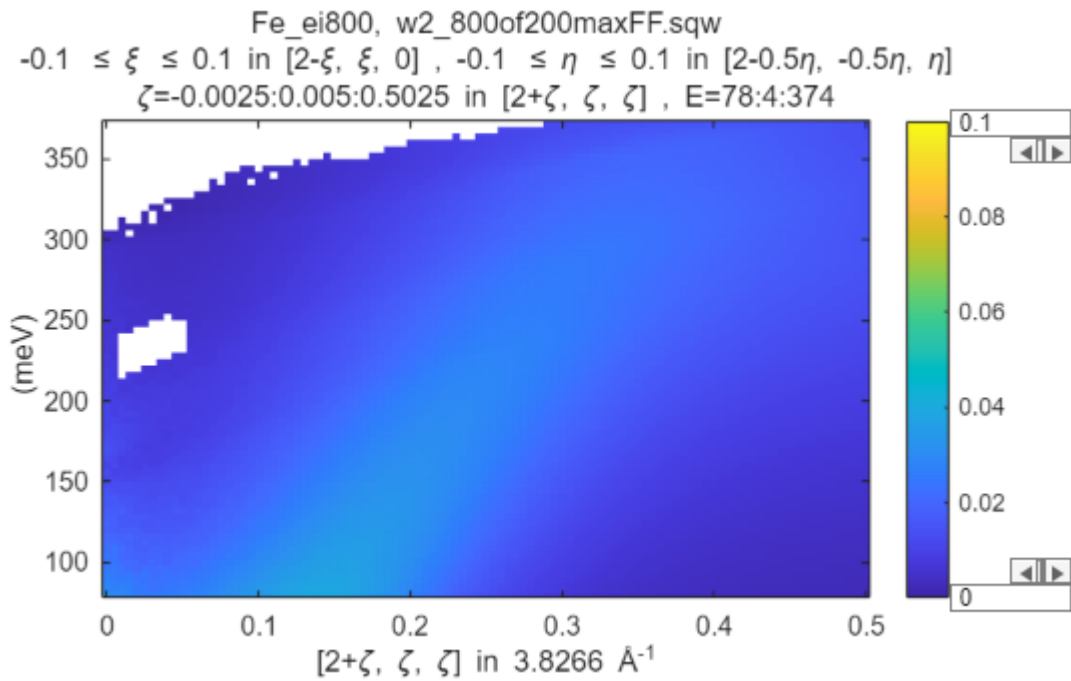


```
data_ranges = cuts_rangesNG.values;
data_ranges = [data_ranges{:}];
pan_data = struct('data_ranges',data_ranges,'combined_ds',w2);
%save('NG_0p5_0p5_0_data','pan_data');
fh.UserData = w2.data;
```

```
plot(spp.Ei800_off200NGmaxNbg); lz 0 0.1;keep_figure;
```

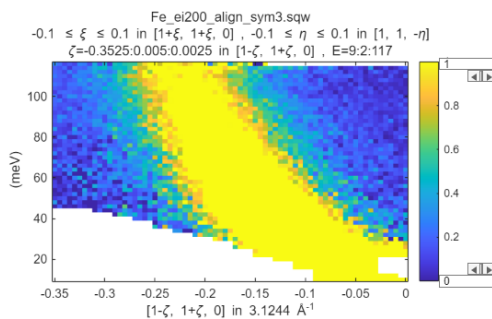


```
plot(fit_obj); lz 0 0.1;keep_figure;
```

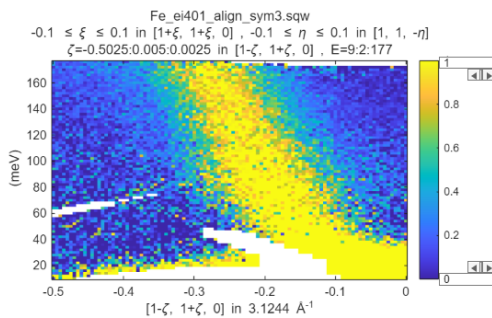


```
%[fit_obj,fit_par] = kk.simulate();
```

```
if ~exist('magi','var')
    magi = MagneticIons();
end
sppCr.Ei200_off110NGminNbg = magi.correct_mag_ff(spp.Ei200_off110NGminNbg);
plot(sppCr.Ei200_off110NGminNbg); lz 0 2;keep_figure;
```

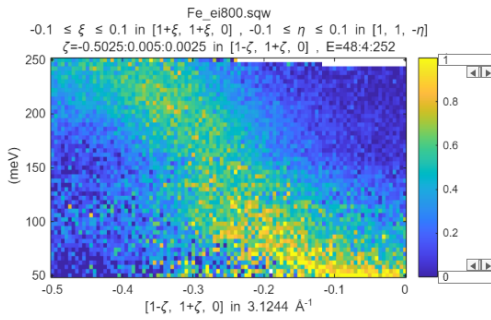


```
sppCr.Ei400_off110NGminNbgCr = magi.correct_mag_ff(spp.Ei400_off110NGminNbg);
plot(sppCr.Ei400_off110NGminNbg); lz 0 1; keep_figure
```

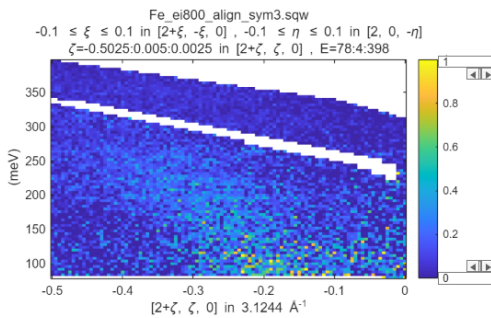


```
sppCr.Ei800_off110NGminNbg = magi.correct_mag_ff(spp.Ei800_off110NGminNbg);
```

```
plot(sppCr.Ei800_off110NGminNbg); lz 0 1; keep_figure
```



```
sppCr.Ei800_off200NGmaxNbg = magi.correct_mag_ff(spp.Ei800_off200NGmaxNbg);  
plot(sppCr.Ei800_off200NGmaxNbg); lz 0 1; keep_figure
```



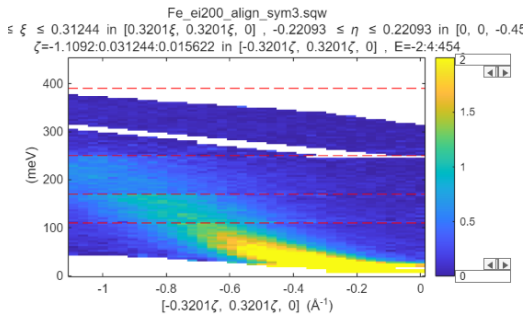
```
[w2,fh] = build_disp_in_dir_with_ds(sppCr,cuts_rangesNG,0.01,0.5,false);
```

```
*** Finished 1of#4 chunks in 0.65649 sec performing sqw_op_bin_pixels
```

```
...
```

```
*** Completed sqw_op_bin_pixels using 4 pages in 1.6275 sec.
```

```
*** Resulting object is backed by file:
```



```
data_ranges = cuts_rangesNG.values;  
data_ranges = [data_ranges{:}];  
pan_data = struct('data_ranges',data_ranges,'combined_ds',w2);  
save('NG_0p5_0p5_0_data','pan_data');
```

```
gamma=10;Seff0=0.7917;  
%J0=39; J1=35; % GP  
J0 = 52; J1 = 20; % NG  
correct_ff = 1;  
gap = 0; %  
init_fg_param = [correct_ff,8,gamma,Seff0, 0, J0, J1, 0, 0, 0];  
%init_fg_params = [coffect_ff,T,gamma,Seff, gap, J0, J1, J2, J3, J4];
```

```
free_sw_param = [0      0, 1  ,1  , 0,    1, 1,  0, 0,  0];
```

```
keys = cuts_rangesNG.keys();
n_cuts = 0;
cut_ranges = struct();
for keyIndex = 1:numel(keys)
    currentKey = keys{keyIndex};
    range = cuts_rangesNG(currentKey);
    min_range = 30;
    if contains(currentKey,'Ei400')
        min_range = 40;
    end
    if contains(currentKey,'Ei800')
        min_range = 70;
    end
    if range(1)<min_range
        range(1) = min_range;
    end
    cut_ranges.(currentKey) = range;
    n_cuts = n_cuts + floor((range(2)-range(1))/10);
end
cuts = cell(1,n_cuts);
cutIdx = 0;
en = zeros(1,n_cuts);
for keyIndex = 1:numel(keys)
    currentKey = keys{keyIndex};
    range = cut_ranges.(currentKey);
    ncc = floor((range(2)-range(1))/10 - 1);
    for i=1:ncc
        cutIdx =cutIdx+1;
        cuts{cutIdx} = cut(spp.(currentKey),[],[range(1)+
(i-1)*10,range(1)+i*10]);
        en(cutIdx) = range(1)+5+10*(i-1);
        %plot(cuts{cutIdx});
    end
end
cuts = cuts(1:cutIdx);
```

```
kk = tobyfit(cuts{:});
kk = kk.set_local_foreground;
kk = kk.set_fun(@sqw_iron);

hkl_proj = spp.Ei800_off200NGmaxNbg.data.proj;
kk = kk.set_pin({init_fg_param,hkl_proj});
kk = kk.set_free(free_sw_param);

kk = kk.set_bind({{6,[6,1]},{7,[7,1]}});

%kk = kk.set_bfun (@linear_bg1D); % set_bfun sets the background functions
%
%bg_param = init_bg_param(valid);
```

```
%kk = kk.set_bpin (bg_param); % initial background constant and gradient
%kk = kk.set_bfree (bfree);
```

```
kk = kk.set_options('list',2);
```

```
[fit_obj,fit_par] = kk.fit();
```

```
Warning: Dataset 20:2 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.9802
Warning: Dataset 22:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 24:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
Warning: Dataset 25:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
Warning: Dataset 26:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
Warning: Dataset 27:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 28:4 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 3.9604
Warning: Dataset 29:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 31:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 32:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
```

```
-----
Beginning fit (max 20 iterations)-----
Starting point
```

```
-----
Total time = 52.0804s    Reduced Chi^2 = 65.1865
Free parameter values:
      10      0.7917      52      20      10      0.7917      10
```

```
-----
Iteration = 1
```

```
-----
Total time = 308.1365s    Reduced Chi^2 = 28.4569    Levenberg-Marquardt = 0.1
Free parameter values:
      62.8      1.751      48.1      20      47.83      1.146      34.82
```

```
-----
Iteration = 2
```

```
-----
Total time = 560.4997s    Reduced Chi^2 = 10.4019    Levenberg-Marquardt = 0.01
Free parameter values:
      134.3      3.231      45.74      20      160      2.913      156.9
```

```
-----
Iteration = 3
```

```
-----
Total time = 814.4295s    Reduced Chi^2 = 7.554    Levenberg-Marquardt = 0.001
Free parameter values:
      31.21      1.936      36.4      20      30.14      1.609      52.02
```

```
-----
Iteration = 4
```

```
-----
Total time = 1064.9219s    Reduced Chi^2 = 18.1239    Levenberg-Marquardt = 0.0001
Free parameter values:
      40.08      1.726      30.84      20      41.7      1.451      49.53
Total time = 1116.5597s    Reduced Chi^2 = 18.0426    Levenberg-Marquardt = 0.001
Free parameter values:
      40.28      1.733      30.91      20      41.9      1.456      49.86
Total time = 1168.159s    Reduced Chi^2 = 17.9074    Levenberg-Marquardt = 0.01
Free parameter values:
      41.8      1.784      31.53      20      43.44      1.495      52.46
Total time = 1219.4847s    Reduced Chi^2 = 3.76    Levenberg-Marquardt = 1
Free parameter values:
      41.94      1.837      35.86      20      42.62      1.508      61.4
```

```

-----
Iteration = 5
-----
Total time = 1476.4097s    Reduced Chi^2 = 2.9607    Levenberg-Marquardt = 0.1
Free parameter values:
    50.9        1.922        33.87        20        55.98        1.636        62.23
-----
Iteration = 6
-----
Total time = 1734.5507s    Reduced Chi^2 = 3.1279    Levenberg-Marquardt = 0.01
Free parameter values:
    47.9        1.863        33.91        20        53.09        1.593        58.52
Total time = 1785.7873s    Reduced Chi^2 = 2.9138    Levenberg-Marquardt = 0.1
Free parameter values:
    48.28       1.872        33.93        20        53.43        1.599        58.93
-----
Iteration = 7
-----
Total time = 2032.9775s    Reduced Chi^2 = 6.1632    Levenberg-Marquardt = 0.01
Free parameter values:
    48.14       1.865        33.78        20        53.11        1.591        58.43
Total time = 2081.5471s    Reduced Chi^2 = 2.9096    Levenberg-Marquardt = 0.1
Free parameter values:
    48.24       1.868        33.8        20        53.24        1.593        58.58
-----
Iteration = 8
-----
Total time = 2336.8084s    Reduced Chi^2 = 2.8944    Levenberg-Marquardt = 0.01
Free parameter values:
    47.8        1.859        33.77        20        52.82        1.588        58.11
-----
Iteration = 9
-----
Total time = 2592.9528s    Reduced Chi^2 = 15.7593    Levenberg-Marquardt = 0.001
Free parameter values:
    47.8        1.859        33.75        20        52.78        1.586        58.05
Total time = 2643.5632s    Reduced Chi^2 = 2.9714    Levenberg-Marquardt = 0.01
Free parameter values:
    47.8        1.859        33.75        20        52.78        1.586        58.06
Total time = 2692.2046s    Reduced Chi^2 = 2.8973    Levenberg-Marquardt = 0.1
Free parameter values:
    47.81       1.859        33.75        20        52.8        1.587        58.07
Total time = 2742.2073s    Reduced Chi^2 = 2.8911    Levenberg-Marquardt = 10
Free parameter values:
    47.81       1.859        33.76        20        52.82        1.587        58.1
-----
Iteration = 10
-----
Total time = 2998.2307s    Reduced Chi^2 = 2.8965    Levenberg-Marquardt = 1
Free parameter values:
    47.78       1.859        33.76        20        52.75        1.587        58.09
Total time = 3049.4974s    Reduced Chi^2 = 2.893    Levenberg-Marquardt = 10
Free parameter values:
    47.8        1.859        33.76        20        52.8        1.587        58.1
Total time = 3100.5366s    Reduced Chi^2 = 2.8919    Levenberg-Marquardt = 100
Free parameter values:
    47.8        1.859        33.76        20        52.81        1.587        58.1
Total time = 3151.9746s    Reduced Chi^2 = 2.895    Levenberg-Marquardt = 10000
Free parameter values:
    47.81       1.859        33.76        20        52.82        1.587        58.1
Total time = 3203.228s    Reduced Chi^2 = 2.8913    Levenberg-Marquardt = 1000000
Free parameter values:
    47.81       1.859        33.76        20        52.82        1.587        58.1
Total time = 3254.7184s    Reduced Chi^2 = 2.8942    Levenberg-Marquardt = 100000000
Free parameter values:
    47.81       1.859        33.76        20        52.82        1.587        58.1
*** No improvement in chi-squared over previous iteration ***
Total time = 3254.7436s    Reduced Chi^2 = 2.8911

```

Free parameter values:

47.81	1.859	33.76	20	52.82	1.587	58.1
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Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

Function 1:

1	1		
2	8		
3	47.81	+/-	Inf
4	1.859	+/-	Inf
5	0		
6	33.76	+/-	Inf
7	20	+/-	Inf
8	0		
9	0		
10	0		

Function 2:

1	1			
2	8			
3	52.82	+/-	Inf	
4	1.587	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 3:

1	1			
2	8			
3	58.1	+/-	Inf	
4	1.49	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 4:

1	1			
2	8			
3	54.99	+/-	Inf	
4	1.294	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 5:

1	1			
2	8			
3	54.29	+/-	Inf	
4	1.215	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			

10	0			
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Function 6:

1	1			
2	8			
3	53.14	+/-	Inf	
4	1.119	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 7:

1	1			
2	8			
3	49.49	+/-	Inf	
4	1.019	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 8:

1	1			
2	8			
3	49.57	+/-	Inf	
4	1.049	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 9:

1	1			
2	8			
3	48.16	+/-	Inf	
4	1.025	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 10:

1	1			
2	8			
3	48.4	+/-	Inf	
4	0.9797	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 11:

1	1			
2	8			
3	48.08	+/-	Inf	
4	0.9793	+/-	Inf	
5	0			

6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 12:

1	1			
2	8			
3	47.19	+/-	Inf	
4	0.9564	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 13:

1	1			
2	8			
3	49.82	+/-	Inf	
4	0.7317	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 14:

1	1			
2	8			
3	51	+/-	Inf	
4	0.7493	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 15:

1	1			
2	8			
3	60.75	+/-	Inf	
4	0.811	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 16:

1	1			
2	8			
3	60.16	+/-	Inf	
4	0.8066	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 17:

1	1
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2	8			
3	59.24	+/-	Inf	
4	0.77	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 18:

1	1			
2	8			
3	66.9	+/-	Inf	
4	0.7802	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 19:

1	1			
2	8			
3	76.45	+/-	Inf	
4	0.7901	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 20:

1	1			
2	8			
3	150.1	+/-	Inf	
4	1.67	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 21:

1	1			
2	8			
3	86.6	+/-	Inf	
4	0.8219	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 22:

1	1			
2	8			
3	59.32	+/-	Inf	
4	0.469	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			

10	0			
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Function 23:

1	1			
2	8			
3	33.76	+/-	Inf	
4	0.3699	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 24:

1	1			
2	8			
3	298.2	+/-	Inf	
4	2.142	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 25:

1	1			
2	8			
3	196.6	+/-	Inf	
4	1.341	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 26:

1	1			
2	8			
3	222.5	+/-	Inf	
4	0.9923	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 27:

1	1			
2	8			
3	214.4	+/-	Inf	
4	1.311	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 28:

1	1			
2	8			
3	2500	+/-	Inf	
4	11.15	+/-	Inf	
5	0			

6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 29:

1	1			
2	8			
3	268.4	+/-	Inf	
4	2.006	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 30:

1	1			
2	8			
3	-187.3	+/-	Inf	
4	1.42	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

Function 31:

1	1			
2	8			
3	114.9	+/-	Inf	
4	0.9534	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

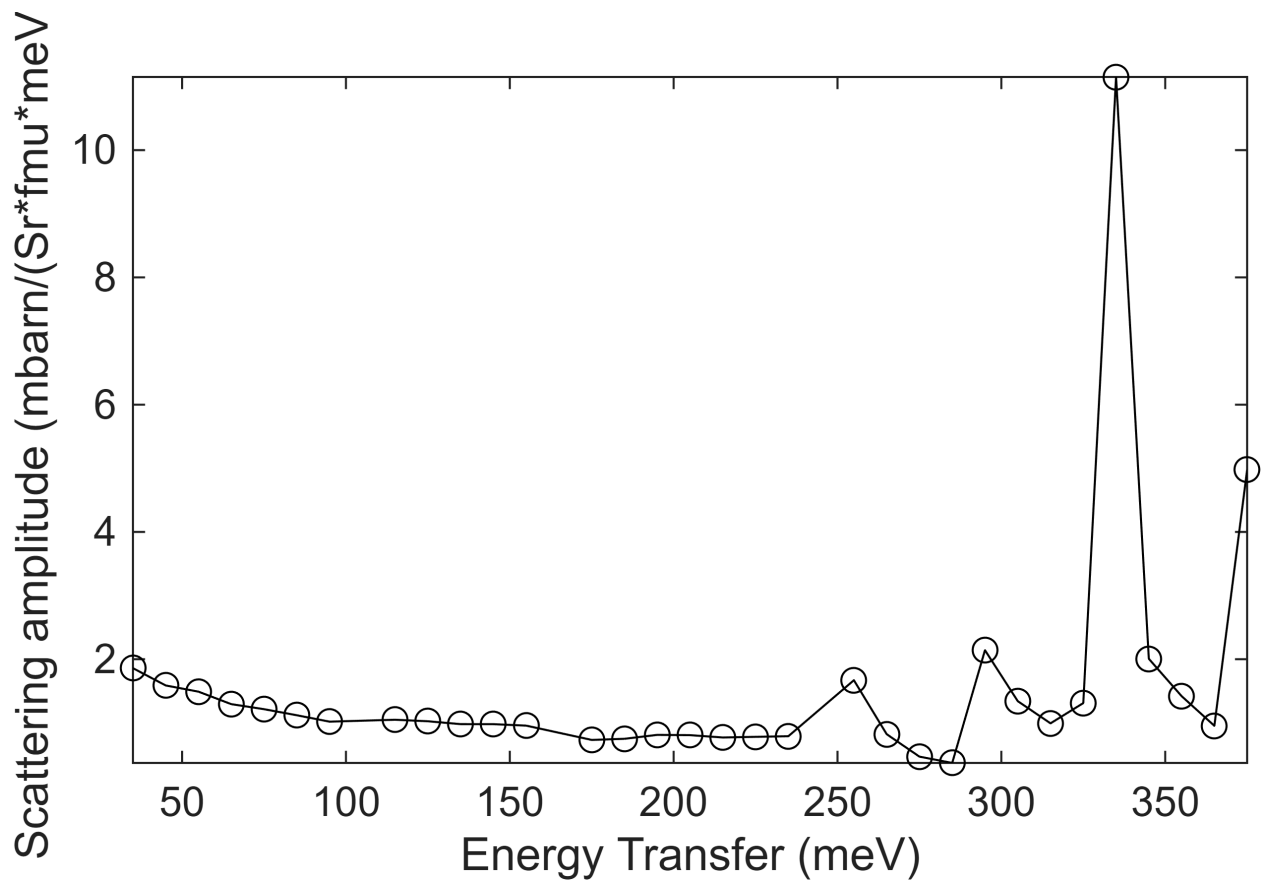
Function 32:

1	1			
2	8			
3	-4.916	+/-	Inf	
4	4.978	+/-	Inf	
5	0			
6	33.76	+/-	Inf	bound to parameter 6 of foreground 1
7	20	+/-	Inf	bound to parameter 7 of foreground 1
8	0			
9	0			
10	0			

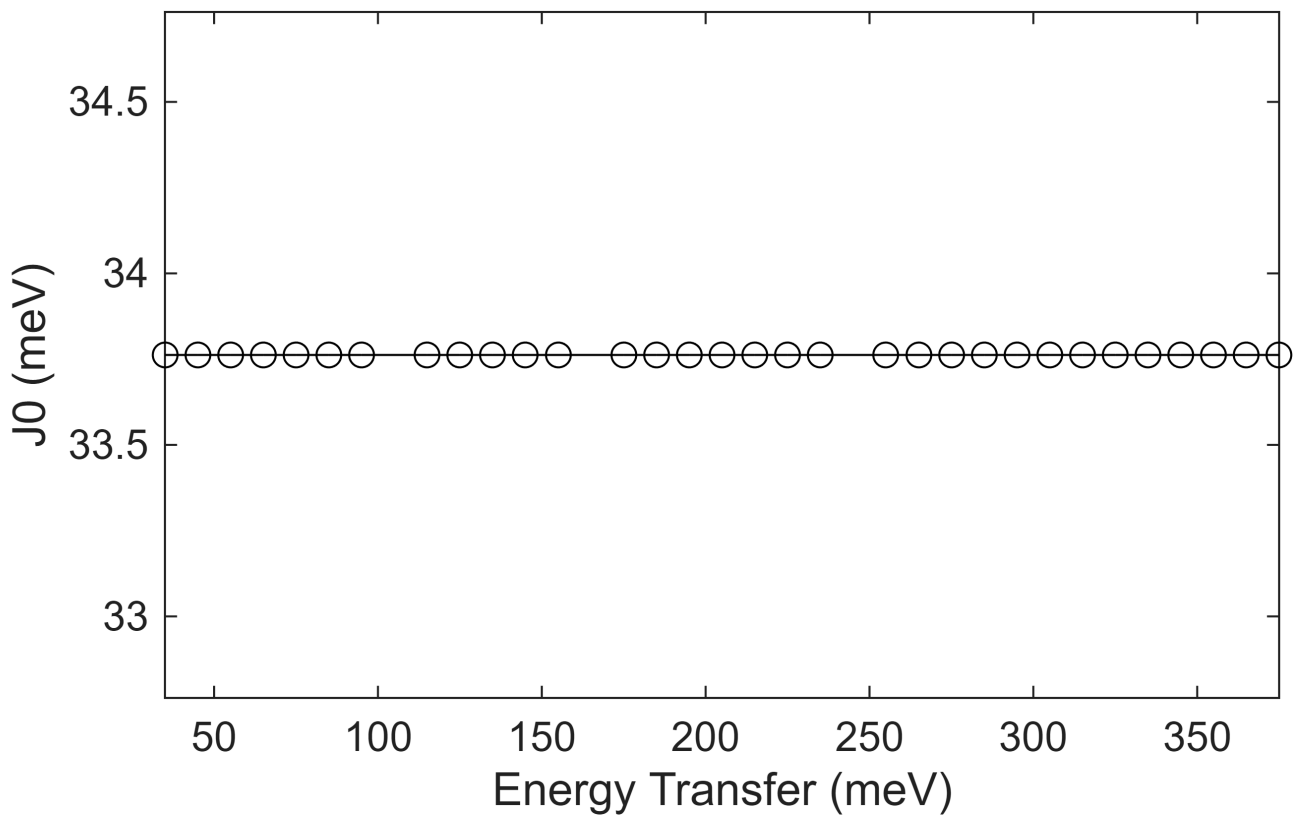
Covariance matrix for free parameters:

NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

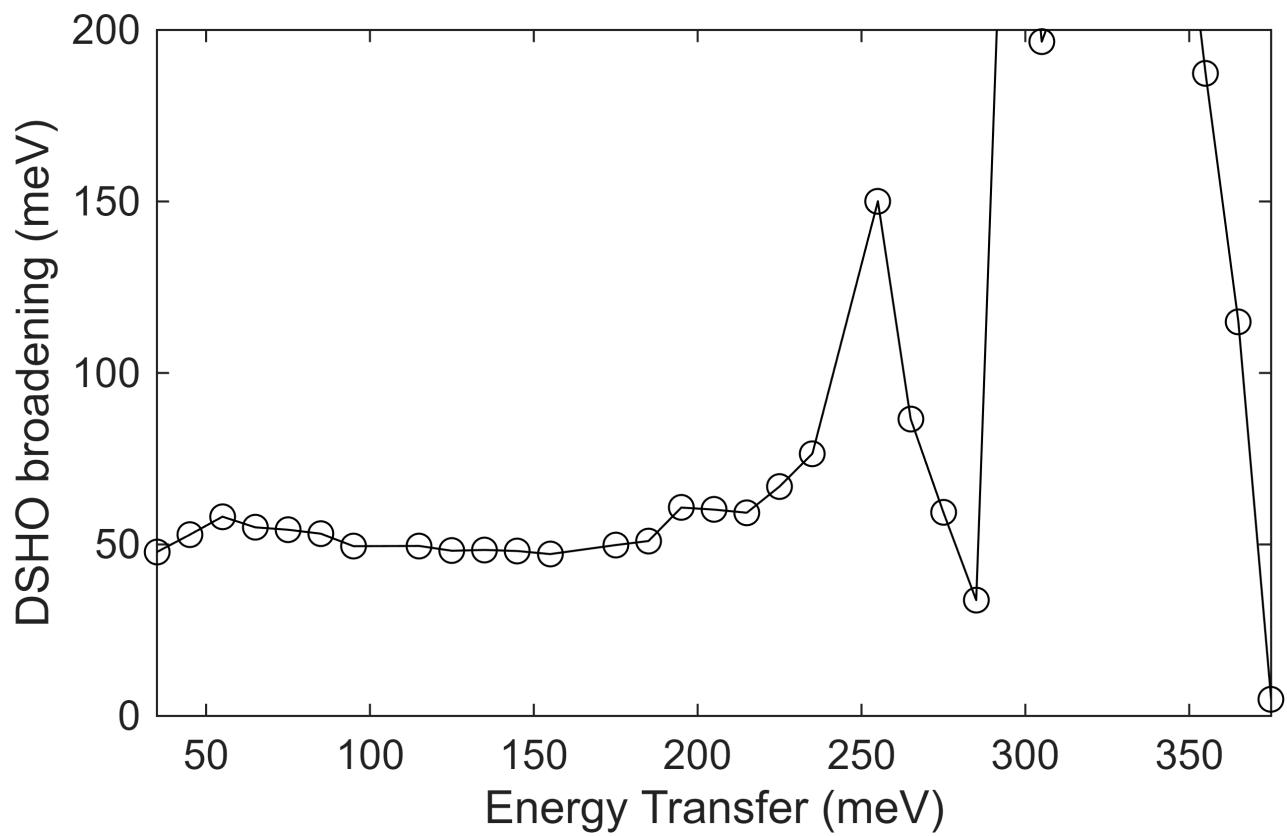
```
[Sar,J0ar,GamAr,J1ar] = extract_multicut_fit_par(fit_par,en);
pd(Sar); keep_figure
```



```
pd(J0ar); keep_figure
```



```
pd(GamAr);ly 0 200; keep_figure;
```



```
pd(J1ar);keep_figure;
```

